ADITYA UNIVERSITY

Freshman Engineering Department
Department of Mathematics
Practice Questions

AY: 2025-26 **SEM-II**

Course Name & Code: Differential Equations and Vector Calculus (2501MA02).

Unit-II: Linear Differential Equations of higher order.

Solve the following Differential Equations (From 1 to 16):

1. $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0$, given that $y(0) = 0, y^{|}(0) = 15$.
2. $\frac{d^4x}{dt^4} + 4x = 0.$
3. $(D^2 - 6D + 9)y = 6e^{3x} + 7e^{-2x} - log2.$
4. $(D^2 + 5D - 6)y = \sin 4x \sin x.$
5. $(D^3 + 4D)y = 5 + sin2x + 3^x.$
6. $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4.$
7. $(D^2 - 4D + 4)y = e^{2x} + 8x^2.$
8. $(D^2 - 1)y = xsin3x + \cos x$.
9. $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = x^2e^{-3x}.$
10. $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x \sin x.$
11. $(D^2 - 4D + 4)y = 8x^2e^{2x}sin2x.$
12. $y^{||} - 6y^{|} + 9y = \frac{e^{3x}}{x^2}$ by the method of variation of parameters.
13. $\frac{d^2y}{dx^2} + a^2y = \tan ax$ by the method of variation of parameters.
14. $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = logx \sin(logx).$
15. $(2x + 3)^2\frac{d^2y}{dx^2} - (2x + 3)\frac{dy}{dx} - 12y = 6x.$
16. Solve the simultaneous equations: $x^{|} + 2y + \sin t = 0,\ y^{|} - 2x - \cos t = 0.$
17. A condenser of capacity C discharged through an inductance L and resistor R in series and the charge q at any time t satisfies the equation, $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = 0$. If $L = 0.25$ henries, $R = 250$ ohms and $C = 2 \times 10^{-6}$ farads. Then find the charge q at any time $t > 0$, given that at initially charge $q = 0.002$ coulombs and current $\frac{dq}{dt}$ is zero in the circuit.

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